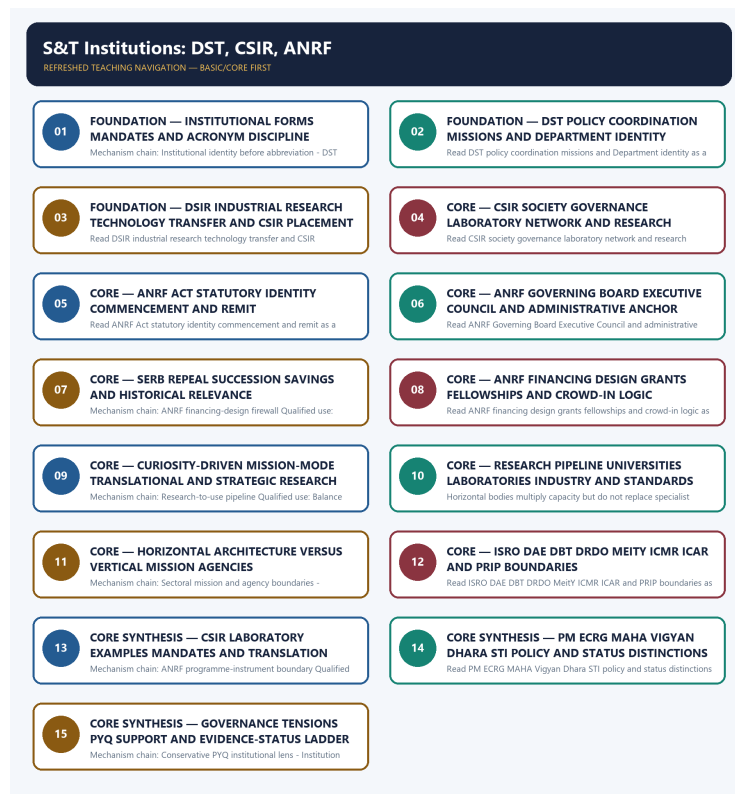


## SOLVED PRACTICE WORKBOOK

# S&T Institutions: DST, CSIR, ANRF - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

# CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

## **S&T Institutions: DST, CSIR, ANRF - Solved Practice Workbook . . . . . 6**

### **BASIC MCQS / REMEDIATION . . . . . 6**

|                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------|----|
| Q1. Which statement correctly identifies Institutional identity before abbreviation? . . . . .                                    | 6  |
| Q2. Which option preserves the technical boundary of Institutional identity before abbreviation? . . . . .                        | 6  |
| Q3. Which statement uses Institutional identity before abbreviation without changing its institution, unit or status? . . . . .   | 7  |
| Q4. Which option avoids the standard UPSC close-option trap about Institutional identity before abbreviation? . . . . .           | 7  |
| Q5. Which statement correctly identifies DST horizontal policy role? . . . . .                                                    | 7  |
| Q6. Which option preserves the technical boundary of DST horizontal policy role? . . . . .                                        | 8  |
| Q7. Which statement uses DST horizontal policy role without changing its institution, unit or status? . . . . .                   | 8  |
| Q8. Which option avoids the standard UPSC close-option trap about DST horizontal policy role? . . . . .                           | 9  |
| Q9. Which statement correctly identifies DSIR industrial-research role? . . . . .                                                 | 9  |
| Q10. Which option preserves the technical boundary of DSIR industrial-research role? . . . . .                                    | 10 |
| Q11. Which statement uses DSIR industrial-research role without changing its institution, unit or status? . . . . .               | 10 |
| Q12. Which option avoids the standard UPSC close-option trap about DSIR industrial-research role? . . . . .                       | 10 |
| Q13. Which statement correctly identifies CSIR legal and research-performing identity? . . . . .                                  | 11 |
| Q14. Which option preserves the technical boundary of CSIR legal and research-performing identity? . . . . .                      | 11 |
| Q15. Which statement uses CSIR legal and research-performing identity without changing its institution, unit or status? . . . . . | 12 |
| Q16. Which option avoids the standard UPSC close-option trap about CSIR legal and research-performing identity? . . . . .         | 12 |
| Q17. Which statement correctly identifies CSIR governance and mandate boundary? . . . . .                                         | 13 |
| Q18. Which option preserves the technical boundary of CSIR governance and mandate boundary? . . . . .                             | 13 |
| Q19. Which statement uses CSIR governance and mandate boundary without changing its institution, unit or status? . . . . .        | 13 |
| Q20. Which option avoids the standard UPSC close-option trap about CSIR governance and mandate boundary? . . . . .                | 14 |
| Q21. Which statement correctly identifies ANRF statutory identity and commencement? . . . . .                                     | 14 |
| Q22. Which option preserves the technical boundary of ANRF statutory identity and commencement? . . . . .                         | 15 |
| Q23. Which statement uses ANRF statutory identity and commencement without changing its institution, unit or status? . . . . .    | 15 |
| Q24. Which option avoids the standard UPSC close-option trap about ANRF statutory identity and commencement? . . . . .            | 15 |
| Q25. Which statement correctly identifies ANRF two-tier governance? . . . . .                                                     | 16 |
| Q26. Which option preserves the technical boundary of ANRF two-tier governance? . . . . .                                         | 16 |
| Q27. Which statement uses ANRF two-tier governance without changing its institution, unit or status? . . . . .                    | 17 |

|                                                                                                                                     |    |
|-------------------------------------------------------------------------------------------------------------------------------------|----|
| Q28. Which option avoids the standard UPSC close-option trap about ANRF two-tier governance? . . . . .                              | 17 |
| Q29. Which statement correctly identifies SERB repeal and succession? . . . . .                                                     | 17 |
| Q30. Which option preserves the technical boundary of SERB repeal and succession? . . . . .                                         | 18 |
| Q31. Which statement uses SERB repeal and succession without changing its institution, unit or status? . . . . .                    | 18 |
| Q32. Which option avoids the standard UPSC close-option trap about SERB repeal and succession? . . . . .                            | 19 |
| Q33. Which statement correctly identifies ANRF financing-design firewall? . . . . .                                                 | 19 |
| Q34. Which option preserves the technical boundary of ANRF financing-design firewall? . . . . .                                     | 19 |
| Q35. Which statement uses ANRF financing-design firewall without changing its institution, unit or status? . . . . .                | 20 |
| Q36. Which option avoids the standard UPSC close-option trap about ANRF financing-design firewall? . . . . .                        | 20 |
| Q37. Which statement correctly identifies Grant fellowship and mission-mode distinction? . . . . .                                  | 21 |
| Q38. Which option preserves the technical boundary of Grant fellowship and mission-mode distinction? . . . . .                      | 21 |
| Q39. Which statement uses Grant fellowship and mission-mode distinction without changing its institution, unit or status? . . . . . | 21 |
| Q40. Which option avoids the standard UPSC close-option trap about Grant fellowship and mission-mode distinction? . . . . .         | 22 |
| Q41. Which statement correctly identifies Research-to-use pipeline? . . . . .                                                       | 22 |
| Q42. Which option preserves the technical boundary of Research-to-use pipeline? . . . . .                                           | 23 |
| Q43. Which statement uses Research-to-use pipeline without changing its institution, unit or status? . . . . .                      | 23 |
| Q44. Which option avoids the standard UPSC close-option trap about Research-to-use pipeline? . . . . .                              | 23 |
| Q45. Which statement correctly identifies Horizontal versus vertical architecture? . . . . .                                        | 24 |
| Q46. Which option preserves the technical boundary of Horizontal versus vertical architecture? . . . . .                            | 24 |
| Q47. Which statement uses Horizontal versus vertical architecture without changing its institution, unit or status? . . . . .       | 25 |
| Q48. Which option avoids the standard UPSC close-option trap about Horizontal versus vertical architecture? . . . . .               | 25 |
| Q49. Which statement correctly identifies Sectoral mission and agency boundaries? . . . . .                                         | 25 |
| Q50. Which option preserves the technical boundary of Sectoral mission and agency boundaries? . . . . .                             | 26 |
| Q51. Which statement uses Sectoral mission and agency boundaries without changing its institution, unit or status? . . . . .        | 26 |
| Q52. Which option avoids the standard UPSC close-option trap about Sectoral mission and agency boundaries? . . . . .                | 27 |
| Q53. Which statement correctly identifies University laboratory industry ecosystem? . . . . .                                       | 27 |
| Q54. Which option preserves the technical boundary of University laboratory industry ecosystem? . . . . .                           | 28 |
| Q55. Which statement uses University laboratory industry ecosystem without changing its institution, unit or status? . . . . .      | 28 |
| Q56. Which option avoids the standard UPSC close-option trap about University laboratory industry ecosystem? . . . . .              | 28 |
| Q57. Which statement correctly identifies Named CSIR laboratory function map? . . . . .                                             | 29 |
| Q58. Which option preserves the technical boundary of Named CSIR laboratory function map? . . . . .                                 | 29 |
| Q59. Which statement uses Named CSIR laboratory function map without changing its institution, unit or status? . . . . .            | 30 |

|                                                                                                                              |    |
|------------------------------------------------------------------------------------------------------------------------------|----|
| Q60. Which option avoids the standard UPSC close-option trap about Named CSIR laboratory function map? . . . . .             | 30 |
| Q61. Which statement correctly identifies ANRF programme-instrument boundary? . . . . .                                      | 30 |
| Q62. Which option preserves the technical boundary of ANRF programme-instrument boundary? . . . . .                          | 31 |
| Q63. Which statement uses ANRF programme-instrument boundary without changing its institution, unit or status? . . . . .     | 31 |
| Q64. Which option avoids the standard UPSC close-option trap about ANRF programme-instrument boundary? . . . . .             | 32 |
| Q65. Which statement correctly identifies STI policy and Vigyan Dhara boundary? . . . . .                                    | 32 |
| Q66. Which option preserves the technical boundary of STI policy and Vigyan Dhara boundary? . . . . .                        | 32 |
| Q67. Which statement uses STI policy and Vigyan Dhara boundary without changing its institution, unit or status? . . . . .   | 33 |
| Q68. Which option avoids the standard UPSC close-option trap about STI policy and Vigyan Dhara boundary? . . . . .           | 33 |
| Q69. Which statement correctly identifies Governance tensions and inclusion? . . . . .                                       | 34 |
| Q70. Which option preserves the technical boundary of Governance tensions and inclusion? . . . . .                           | 34 |
| Q71. Which statement uses Governance tensions and inclusion without changing its institution, unit or status? . . . . .      | 34 |
| Q72. Which option avoids the standard UPSC close-option trap about Governance tensions and inclusion? . . . . .              | 35 |
| Q73. Which statement correctly identifies Conservative PYQ institutional lens? . . . . .                                     | 35 |
| Q74. Which option preserves the technical boundary of Conservative PYQ institutional lens? . . . . .                         | 36 |
| Q75. Which statement uses Conservative PYQ institutional lens without changing its institution, unit or status? . . . . .    | 36 |
| Q76. Which option avoids the standard UPSC close-option trap about Conservative PYQ institutional lens? . . . . .            | 36 |
| Q77. Which statement correctly identifies Institution and status evidence ladder? . . . . .                                  | 37 |
| Q78. Which option preserves the technical boundary of Institution and status evidence ladder? . . . . .                      | 37 |
| Q79. Which statement uses Institution and status evidence ladder without changing its institution, unit or status? . . . . . | 38 |
| Q80. Which option avoids the standard UPSC close-option trap about Institution and status evidence ladder? . . . . .         | 38 |
| Semantic-completeness coverage drills - Topic 24 . . . . .                                                                   | 39 |

## **PYQS AND ANSWER PRACTICE . . . . . 40**

|                                                    |    |
|----------------------------------------------------|----|
| AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP . . . . .     | 40 |
| PYQ DEMAND CARD 1 - 2018 and 2021 GS-III . . . . . | 40 |
| PYQ DEMAND CARD 2 - 2018 and 2021 GS-III . . . . . | 41 |
| PYQ DEMAND CARD 3 - 2025 GS-III . . . . .          | 43 |
| ORIGINAL MAINS 1 - 10 MARKS . . . . .              | 44 |
| ORIGINAL MAINS 2 - 10 MARKS . . . . .              | 49 |
| ORIGINAL MAINS 3 - 15 MARKS . . . . .              | 52 |
| ORIGINAL MAINS 4 - 15 MARKS . . . . .              | 55 |
| ORIGINAL MAINS 5 - 20 MARKS . . . . .              | 58 |



# S&T Institutions: DST, CSIR, ANRF - Solved Practice Workbook

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly identifies Institutional identity before abbreviation?

A. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST.

**Answer: A. Explanation: India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q2. Which option preserves the technical boundary of Institutional identity before abbreviation?

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. C. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. D. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.

**Answer: B. Explanation: India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q3. Which statement uses Institutional identity before abbreviation without changing its institution, unit or status?

A. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. B. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. C. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. D. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.

**Answer: C. Explanation: India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q4. Which option avoids the standard UPSC close-option trap about Institutional identity before abbreviation?

A. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. B. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. C. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. D. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers.

**Answer: D. Explanation: India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q5. Which statement correctly identifies DST horizontal policy role?

A. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. B. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a



network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST.

**Answer: A. Explanation: The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q6. Which option preserves the technical boundary of DST horizontal policy role?**

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.

**Answer: B. Explanation: The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q7. Which statement uses DST horizontal policy role without changing its institution, unit or status?**

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. D. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.

**Answer: C. Explanation: The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not**



**India's single all-purpose laboratory network or a Ministry in its own right. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q8. Which option avoids the standard UPSC close-option trap about DST horizontal policy role?**

A. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. B. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. C. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. D. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Answer: D. Explanation: The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q9. Which statement correctly identifies DSIR industrial-research role?**

A. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. B. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council.

**Answer: A. Explanation: The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q10. Which option preserves the technical boundary of DSIR industrial-research role?

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. C. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. D. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.

**Answer: B. Explanation: The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q11. Which statement uses DSIR industrial-research role without changing its institution, unit or status?

A. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: C. Explanation: The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q12. Which option avoids the standard UPSC close-option trap about DSIR industrial-research role?

A. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. B. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. C. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation,

receipt, annual release, grant award, expenditure or research outcome. D. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST.

**Answer: D. Explanation: The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q13. Which statement correctly identifies CSIR legal and research-performing identity?

A. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. D. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.

**Answer: A. Explanation: The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q14. Which option preserves the technical boundary of CSIR legal and research-performing identity?

A. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. B. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. C. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: B. Explanation: The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation,**

environment and information technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

**Q15. Which statement uses CSIR legal and research-performing identity without changing its institution, unit or status?**

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers.

**Answer: C. Explanation: The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q16. Which option avoids the standard UPSC close-option trap about CSIR legal and research-performing identity?**

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. D. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.

**Answer: D. Explanation: The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q17. Which statement correctly identifies CSIR governance and mandate boundary?

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: A. Explanation: The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q18. Which option preserves the technical boundary of CSIR governance and mandate boundary?

A. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. B. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. C. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: B. Explanation: The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q19. Which statement uses CSIR governance and mandate boundary without changing its institution, unit or status?

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to



their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: C. Explanation: The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q20. Which option avoids the standard UPSC close-option trap about CSIR governance and mandate boundary?**

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council.

**Answer: D. Explanation: The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q21. Which statement correctly identifies ANRF statutory identity and commencement?**

A. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. D. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome.

**Answer: A. Explanation: The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q22. Which option preserves the technical boundary of ANRF statutory identity and commencement?**

A. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. B. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. C. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: B. Explanation: The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q23. Which statement uses ANRF statutory identity and commencement without changing its institution, unit or status?**

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. D. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung.

**Answer: C. Explanation: The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q24. Which option avoids the standard UPSC close-option trap about ANRF statutory identity and commencement?**

A. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. B. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. C. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply



capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. D. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.

**Answer: D. Explanation: The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q25. Which statement correctly identifies ANRF two-tier governance?**

A. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. B. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. C. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: A. Explanation: Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q26. Which option preserves the technical boundary of ANRF two-tier governance?**

A. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. B. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. C. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: B. Explanation: Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q27. Which statement uses ANRF two-tier governance without changing its institution, unit or status?**

A. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. B. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. C. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: C. Explanation: Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q28. Which option avoids the standard UPSC close-option trap about ANRF two-tier governance?**

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers.

**Answer: D. Explanation: Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q29. Which statement correctly identifies SERB repeal and succession?**

A. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation,

manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome.

**Answer: A. Explanation: The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q30. Which option preserves the technical boundary of SERB repeal and succession?

A. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. B. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: B. Explanation: The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q31. Which statement uses SERB repeal and succession without changing its institution, unit or status?

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. D. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung.

**Answer: C. Explanation: The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q32. Which option avoids the standard UPSC close-option trap about SERB repeal and succession?

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Answer: D. Explanation: The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q33. Which statement correctly identifies ANRF financing-design firewall?

A. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: A. Explanation: The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q34. Which option preserves the technical boundary of ANRF financing-design firewall?

A. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. B. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. C. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. D. DST, ANRF and



DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards.

**Answer: B. Explanation: The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q35. Which statement uses ANRF financing-design firewall without changing its institution, unit or status?**

A. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. D. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.

**Answer: C. Explanation: The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q36. Which option avoids the standard UPSC close-option trap about ANRF financing-design firewall?**

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. C. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. D. The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome.

**Answer: D. Explanation: The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q37. Which statement correctly identifies Grant fellowship and mission-mode distinction?

A. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. B. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. C. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. D. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung.

**Answer: A. Explanation: A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q38. Which option preserves the technical boundary of Grant fellowship and mission-mode distinction?

A. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. B. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.

**Answer: B. Explanation: A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q39. Which statement uses Grant fellowship and mission-mode distinction without changing its institution, unit or status?

A. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. B. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. C. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different



risk and time horizons; neither can substitute completely for the other. D. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.

**Answer: C. Explanation: A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q40. Which option avoids the standard UPSC close-option trap about Grant fellowship and mission-mode distinction?**

A. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. B. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Answer: D. Explanation: A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q41. Which statement correctly identifies Research-to-use pipeline?**

A. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. B. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. C. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. D. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards.

**Answer: A. Explanation: A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q42. Which option preserves the technical boundary of Research-to-use pipeline?

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field.

**Answer: B. Explanation: A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q43. Which statement uses Research-to-use pipeline without changing its institution, unit or status?

A. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. B. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. C. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. D. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems.

**Answer: C. Explanation: A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q44. Which option avoids the standard UPSC close-option trap about Research-to-use pipeline?

A. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. B. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or

drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung.

**Answer: D. Explanation: A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q45. Which statement correctly identifies Horizontal versus vertical architecture?**

A. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. B. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. C. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. D. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.

**Answer: A. Explanation: DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q46. Which option preserves the technical boundary of Horizontal versus vertical architecture?**

A. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. B. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.

**Answer: B. Explanation: DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q47. Which statement uses Horizontal versus vertical architecture without changing its institution, unit or status?**

A. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. D. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.

**Answer: C. Explanation: DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q48. Which option avoids the standard UPSC close-option trap about Horizontal versus vertical architecture?**

A. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards.

**Answer: D. Explanation: DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q49. Which statement correctly identifies Sectoral mission and agency boundaries?**

A. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. B. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research



administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field.

**Answer: A. Explanation: ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q50. Which option preserves the technical boundary of Sectoral mission and agency boundaries?**

A. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. B. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments.

**Answer: B. Explanation: ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q51. Which statement uses Sectoral mission and agency boundaries without changing its institution, unit or status?**

A. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. D. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.



**Answer: C. Explanation: ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q52. Which option avoids the standard UPSC close-option trap about Sectoral mission and agency boundaries?**

A. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. B. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. C. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. D. ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.

**Answer: D. Explanation: ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q53. Which statement correctly identifies University laboratory industry ecosystem?**

A. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field.

**Answer: A. Explanation: A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q54. Which option preserves the technical boundary of University laboratory industry ecosystem?

A. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. B. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. C. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. D. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.

**Answer: B. Explanation: A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q55. Which statement uses University laboratory industry ecosystem without changing its institution, unit or status?

A. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. D. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Answer: C. Explanation: A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q56. Which option avoids the standard UPSC close-option trap about University laboratory industry ecosystem?

A. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. B. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. C. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national

priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. D. A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems.

**Answer: D. Explanation: A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q57. Which statement correctly identifies Named CSIR laboratory function map?

A. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Answer: A. Explanation: CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q58. Which option preserves the technical boundary of Named CSIR laboratory function map?

A. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. B. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. C. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. D. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments.

**Answer: B. Explanation: CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q59. Which statement uses Named CSIR laboratory function map without changing its institution, unit or status?**

A. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. B. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. C. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. D. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology.

**Answer: C. Explanation: CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q60. Which option avoids the standard UPSC close-option trap about Named CSIR laboratory function map?**

A. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. B. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. C. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. D. CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field.

**Answer: D. Explanation: CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q61. Which statement correctly identifies ANRF programme-instrument boundary?**

A. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist



science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. D. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Answer: A. Explanation: The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q62. Which option preserves the technical boundary of ANRF programme-instrument boundary?**

A. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. B. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. C. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. D. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Answer: B. Explanation: The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q63. Which statement uses ANRF programme-instrument boundary without changing its institution, unit or status?**

A. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. B. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. C. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: C. Explanation: The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q64. Which option avoids the standard UPSC close-option trap about ANRF programme-instrument boundary?**

A. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. B. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. C. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. D. The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.

**Answer: D. Explanation: The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q65. Which statement correctly identifies STI policy and Vigyan Dhara boundary?**

A. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. B. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. C. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: A. Explanation: DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q66. Which option preserves the technical boundary of STI policy and Vigyan Dhara boundary?**

A. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. B. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. C. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. D. Constitution by law, notification,

board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: B. Explanation: DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q67. Which statement uses STI policy and Vigyan Dhara boundary without changing its institution, unit or status?**

A. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: C. Explanation: DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q68. Which option avoids the standard UPSC close-option trap about STI policy and Vigyan Dhara boundary?**

A. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. B. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. C. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. D. DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments.

**Answer: D. Explanation: DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q69. Which statement correctly identifies Governance tensions and inclusion?

A. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. B. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. C. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: A. Explanation: Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q70. Which option preserves the technical boundary of Governance tensions and inclusion?

A. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. B. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. C. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. D. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Answer: B. Explanation: Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q71. Which statement uses Governance tensions and inclusion without changing its institution, unit or status?

A. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. D. India's science architecture contains Ministries, Departments,

statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers.

**Answer: C. Explanation: Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q72. Which option avoids the standard UPSC close-option trap about Governance tensions and inclusion?**

A. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. B. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. C. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. D. Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Answer: D. Explanation: Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q73. Which statement correctly identifies Conservative PYQ institutional lens?**

A. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: A. Explanation: No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q74. Which option preserves the technical boundary of Conservative PYQ institutional lens?

A. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. B. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. C. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. D. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Answer: B. Explanation: No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q75. Which statement uses Conservative PYQ institutional lens without changing its institution, unit or status?

A. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. B. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. C. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. D. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Answer: C. Explanation: No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q76. Which option avoids the standard UPSC close-option trap about Conservative PYQ institutional lens?

A. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. B. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. C. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research,



technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. D. No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology.

**Answer: D. Explanation: No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q77. Which statement correctly identifies Institution and status evidence ladder?**

A. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. B. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. C. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. D. India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers.

**Answer: A. Explanation: Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q78. Which option preserves the technical boundary of Institution and status evidence ladder?**

A. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. B. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. C. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. D. The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Answer: B. Explanation: Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q79. Which statement uses Institution and status evidence ladder without changing its institution, unit or status?**

A. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. B. The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. C. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. D. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.

**Answer: C. Explanation: Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q80. Which option avoids the standard UPSC close-option trap about Institution and status evidence ladder?**

A. The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. B. The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council. C. The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. D. Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Answer: D. Explanation: Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Semantic-completeness coverage drills - Topic 24

| Drill | Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                       | Minimum answer route                                                                                          | Fatal trap                             |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------|
| A     | The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. |                                                                                                               |                                        |
| B     | The Prime Minister is President of CSIR and the Union Science and Technology Minister its Vice-President, while laboratories perform research, technology development, testing, standards-related work and translation according to their mandates. High-level office-bearers do not convert CSIR into a Department, regulator or primarily grant-disbursing council.                                                                        |                                                                                                               |                                        |
| C     | The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.                                                             |                                                                                                               |                                        |
| D     | Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.                                                                |                                                                                                               |                                        |
| E     | How should lost schools be reconstructed?                                                                                                                                                                                                                                                                                                                                                                                                    | opponent report -> rare material anchor -> corroboration -> graded confidence                                 | quoting Charvaka as intact scripture   |
| F     | What sustained debate?                                                                                                                                                                                                                                                                                                                                                                                                                       | teacher -> court/&#8203;assembly -> monastery/&#8203;temple -> patron -> commentary                           | philosophy as private timeless insight |
| G     | What can Gargi and Maitreyi prove?                                                                                                                                                                                                                                                                                                                                                                                                           | named textual voice -> debate context -> male archive and access limit                                        | universal equality                     |
| H     | What is the Science and Technology cutoff?                                                                                                                                                                                                                                                                                                                                                                                                   | Gupta/&#8203;post-Gupta scholasticism -> Kumarila/&#8203;Dharmakirti/&#8203;Shankara boundary -> later owners | importing mature medieval Vedanta      |

**PYQ self-check:** separately solve 2024 Q58 through parable plus relative chronology, and 2022 Q56 through Aryadeva-Dignaga-Nathamuni identification.

## PYQS AND ANSWER PRACTICE

### AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

No audited 2018-2026 PYQ is directly routed to Topic 24. Three representative cross-owner cards use biotechnology R&D, physics-discovery impact and semiconductor-mission demands only to demonstrate the supporting funding, laboratory, university and translation lens; direct ownership remains with Topics 13, 21 and 11 respectively.

### PYQ DEMAND CARD 1 - 2018 and 2021 GS-III

**Demand:** Use only the institutional-support dimension of the routed biotechnology and applied-biotechnology R&D demands.

**Status:** Cross-owner supporting card: substantive biotechnology ownership remains with Topic 13 and related specialist owners; this card contributes funding, laboratory, university and translation architecture without claiming the technology answer.

**Model solution: Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Named CSIR laboratory function map:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2018 and 2021 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### Detailed examiner-grade model answer:

**Introduction and thesis: Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Named CSIR laboratory function map:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

#### Analytical body:

1. **Claim and named evidence:** Demand: Use only the institutional-support dimension of the routed biotechnology and applied-biotechnology R&D demands. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Named CSIR laboratory function map:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 1 - 2018 and 2021 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## PYQ DEMAND CARD 2 - 2018 and 2021 GS-III

**Demand:** Use only the research-ecosystem and translation dimension of the Bose-Einstein-statistics and blue-LED impact demands.

**Status:** Cross-owner supporting card: Topic 21 retains the physics and device answer; Topic 24 supplies the distinction among long-horizon research, laboratory capability, validation, scale and public outcome.

**Model solution: Grant fellowship and mission-mode distinction:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Conservative PYQ institutional lens:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may



supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 2 - 2018 and 2021 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Grant fellowship and mission-mode distinction:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Conservative PYQ institutional lens:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Analytical body:**

1. **Claim and named evidence:** Demand: Use only the research-ecosystem and translation dimension of the Bose-Einstein-statistics and blue-LED impact demands. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Cross-owner supporting card: Topic 21 retains the physics and device answer; Topic 24 supplies the distinction among long-horizon research, laboratory capability, validation, scale and public outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Grant fellowship and mission-mode distinction:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Research-to-use pipeline:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Conservative PYQ institutional lens:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. The solution therefore answers the routed demand without

inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For “PYQ DEMAND CARD 2 - 2018 and 2021 GS-III”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

### PYQ DEMAND CARD 3 - 2025 GS-III

**Demand:** Use only the horizontal research-architecture dimension of the semiconductor-industry and India Semiconductor Mission demand.

**Status:** Cross-owner supporting card: Topic 11 retains direct ownership; this card adds universities, ANRF, DST, CSIR, skills and translation constraints without rewriting a mission announcement as manufacturing success.

**Model solution: Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Sectoral mission and agency boundaries:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Institution and status evidence ladder:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 3 - 2025 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### Detailed examiner-grade model answer:

**Introduction and thesis: Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Sectoral mission and agency boundaries:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Institution and status evidence ladder:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### Analytical body:

1. **Claim and named evidence:** Demand: Use only the horizontal research-architecture dimension of the semiconductor-industry and India Semiconductor Mission demand. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Cross-owner supporting card: Topic 11 retains direct ownership; this card adds universities, ANRF, DST, CSIR, skills and translation constraints without rewriting a mission announcement as manufacturing success. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Horizontal versus vertical architecture:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Sectoral mission and agency boundaries:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **University laboratory industry ecosystem:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Institution and status evidence ladder:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 3 - 2025 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

### ORIGINAL MAINS 1 - 10 MARKS

**Question:** Distinguish DST, DSIR, CSIR, ANRF and SERB by legal identity, parentage and function. Answer in about 150 words.

**Model thesis: Claim:** Institutional identity before abbreviation. **Named evidence/example:** India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and

technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DSIR industrial-research role. **Named evidence/example:** The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SERB repeal and succession. **Named evidence/example:** The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers.
- The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.
- The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST.
- The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.
- The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.



- The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity.

**Qualified conclusion: Claim:** Institutional identity before abbreviation. **Named evidence/example:** India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DSIR industrial-research role. **Named evidence/example:** The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SERB repeal and succession. **Named evidence/example:** The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **answer** requires a direct position on "Distinguish DST, DSIR, CSIR, ANRF and SERB by legal identity, parentage and function. Answer...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Institutional identity before abbreviation. **Named evidence/example:** India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. **Analysis:**



This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DSIR industrial-research role. **Named evidence/example:** The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SERB repeal and succession. **Named evidence/example:** The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant

council or another name for DST. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Institutional identity before abbreviation. **Named evidence/example:** India's science architecture contains Ministries, Departments, statutory bodies, autonomous societies, offices, academies, regulators, grant programmes and research-performing laboratories. Legal form determines accountability and powers, while mandate determines function; an acronym or shared ministry does not make two bodies institutional peers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DSIR industrial-research role. **Named evidence/example:** The Department of Scientific and Industrial Research is a separate Department in the Ministry of Science and Technology; it promotes industrial research, technology transfer and recognition or support mechanisms for industrial R&D and administratively holds CSIR. DSIR is not a laboratory, a generic grant council or another name for DST. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal

character and development-to-deployment status boundary. **Claim:** ANRF statutory identity and commencement.

**Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SERB repeal and succession. **Named evidence/example:** The Science and Engineering Research Board was the statutory competitive-grants body under the SERB Act, 2008; section 27 of the ANRF Act repealed that Act and dissolved SERB with transition and savings for assets, liabilities and personnel. SERB remains historically relevant in older schemes and PYQs but is not a separate continuing statutory apex identity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Distinguish DST, DSIR, CSIR, ANRF and SERB by legal identity, parentage and function. Answer...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 2 - 10 MARKS

**Question:** Explain ANRF's statutory governance, financing design and research instruments. Answer in about 150 words.

**Model thesis:** **Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF two-tier governance. **Named evidence/example:** Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution,

application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office.
- Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers.
- The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome.
- A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.

**Qualified conclusion: Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF two-tier governance. **Named evidence/example:** Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **explain** requires a direct position on "Explain ANRF's statutory governance, financing design and research instruments. Answer in...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**



**Introduction and thesis: Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF two-tier governance. **Named evidence/example:** Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:**



State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** ANRF statutory identity and commencement. **Named evidence/example:** The Anusandhan National Research Foundation is a statutory apex research body created by the ANRF Act, 2023, Act 25 of 2023, enacted on 12 August 2023 and brought into force on 5 February 2024. It is designed to seed, grow and promote research and provide high-level strategic direction; DST is the administrative anchor, not a basis for calling ANRF a DST sub-office. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF two-tier governance. **Named evidence/example:** Section 5 places the Prime Minister as ex officio President of ANRF's Governing Board, with the Union Ministers for Science and Technology and Education as Vice-Presidents, while section 7 makes the Principal Scientific Adviser ex officio Chairperson of the Executive Council. Strategic governance, executive implementation and administrative support are therefore distinct layers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Explain ANRF's statutory governance, financing design and research instruments. Answer in...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

### ORIGINAL MAINS 3 - 15 MARKS

**Question:** Analyse the research-to-use pipeline and the complementary roles of funders, universities, laboratories and industry. Answer in about 250 words.

**Model thesis: Claim:** Research-to-use pipeline. **Named evidence/example:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** University laboratory

industry ecosystem. **Named evidence/example:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Named CSIR laboratory function map. **Named evidence/example:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung.
- A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems.
- CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field.

**Qualified conclusion:** **Claim:** Research-to-use pipeline. **Named evidence/example:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** University laboratory industry ecosystem. **Named evidence/example:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Named CSIR laboratory function map. **Named evidence/example:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **analyse** requires a direct position on "Analyse the research-to-use pipeline and the complementary roles of funders, universities,...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

### Detailed examiner-grade model answer:

**Introduction and thesis:** **Claim:** Research-to-use pipeline. **Named evidence/example:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** University laboratory industry ecosystem. **Named evidence/example:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Named CSIR laboratory function map. **Named evidence/example:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

### Analytical body:

1. **Claim and named evidence:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Research-to-use pipeline. **Named evidence/example:** A disciplined research pipeline separates question formulation, basic research, experimental validation, replication, applied research, prototype, testing, scale-up, standards or regulation, manufacturing, adoption and verified societal outcome. ANRF can fund or shape parts of this pipeline, while CSIR and universities can perform research, but no institutional label proves every later rung. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability

rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** University laboratory industry ecosystem. **Named evidence/example:** A research ecosystem links universities, national laboratories, mission agencies, startups, industry, standards bodies, funders and skilled people. Funding volume alone cannot repair weak proposal support, laboratory access, peer review, procurement, research administration, mentorship, interdisciplinary coordination or absorption by firms and public systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Named CSIR laboratory function map. **Named evidence/example:** CSIR-NPL anchors metrology and measurement science, CSIR-NCL chemistry, CSIR-CCMB cellular and molecular biology, CSIR-IGIB genomics and integrative biology, CSIR-CEERI electronics and CSIR-CECRI electrochemistry. A laboratory example should illustrate a mandate, not be converted into ownership of every mission, product, standard or regulatory decision in that field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Analyse the research-to-use pipeline and the complementary roles of funders, universities,...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** Discuss how horizontal S&T architecture supports but does not replace vertical mission agencies. Answer in about 250 words.

**Model thesis: Claim:** Horizontal versus vertical architecture. **Named evidence/example:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sectoral mission and agency boundaries. **Named evidence/example:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Conservative PYQ institutional lens. **Named evidence/example:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**



- DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards.
- ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category.
- No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology.

**Qualified conclusion: Claim:** Horizontal versus vertical architecture. **Named evidence/example:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sectoral mission and agency boundaries. **Named evidence/example:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Conservative PYQ institutional lens. **Named evidence/example:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **discuss** requires a direct position on "Discuss how horizontal S&T architecture supports but does not replace vertical mission...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Horizontal versus vertical architecture. **Named evidence/example:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sectoral mission and agency boundaries. **Named evidence/example:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Conservative PYQ institutional lens. **Named evidence/example:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of



the underlying discovery or technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Horizontal versus vertical architecture. **Named evidence/example:** DST, ANRF and DSIR-CSIR provide horizontal policy, funding or research capability across sectors, while vertical institutions execute domain mandates. Horizontal bodies multiply capacity but do not replace specialist mission agencies; a vertical mission cannot by itself substitute for broad universities, labs, grants, skills and standards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sectoral mission and agency boundaries. **Named evidence/example:** ISRO is an organisation under the Department of Space; DAE and DBT are Departments; DRDO operates under the Ministry of Defence; MeitY is a Ministry; ICMR is under the Ministry of Health and Family Welfare; ICAR is under DARE; and PRIP belongs to the Department of Pharmaceuticals. These bodies may collaborate with DST, ANRF or CSIR without moving into one common legal category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Conservative PYQ institutional lens. **Named evidence/example:** No audited 2018-2026 question is directly routed to Topic 24; relevant biotechnology, blue-LED, Bose-Einstein and semiconductor demands remain owned by their specialist science topics. Topic 24 may supply only the horizontal funding, laboratory, university and translation dimension and must not claim substantive ownership of the underlying discovery or technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For “Discuss how horizontal S&T architecture supports but does not replace vertical mission...”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Critically examine ANRF as a research-governance reform through autonomy, inclusion, mission balance and private participation. Answer in about 300 words.

**Model thesis: Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF programme-instrument boundary. **Named evidence/example:** The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Governance tensions and inclusion. **Named evidence/example:** Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome.
- A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other.
- The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses.
- Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and

concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs.

**Qualified conclusion: Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF programme-instrument boundary. **Named evidence/example:** The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Governance tensions and inclusion. **Named evidence/example:** Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **critically examine** requires a direct position on "Critically examine ANRF as a research-governance reform through autonomy, inclusion, mission...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF programme-instrument boundary. **Named evidence/example:** The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and

development-to-deployment status boundary. **Claim:** Governance tensions and inclusion. **Named evidence/example:** Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed project, validated technology and public outcome are separate statuses. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** ANRF financing-design firewall. **Named evidence/example:** The owner records an announced ANRF design of 50,000 crore rupees over 2023-28, with about 36,000 crore expected from non-government sources including industry and philanthropy. This is a financing design and crowd-in ambition, not proof of appropriation, receipt, annual release, grant award, expenditure or research outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Grant fellowship and mission-mode distinction. **Named evidence/example:** A research grant funds a defined project, a fellowship principally supports a researcher, and a mission-mode programme directs multi-institutional work toward priority outcomes and milestones. Investigator-led curiosity research and directed strategic research serve different risk and time horizons; neither can substitute completely for the other. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ANRF programme-instrument boundary. **Named evidence/example:** The owners identify PM ECRG as early-career project support and MAHA as a mission-mode platform, with EV mobility and later water or drones calls illustrating directed research. A call for proposals, eligibility rule, selected proposal, sanctioned grant, released instalment, completed



project, validated technology and public outcome are separate statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Governance tensions and inclusion. **Named**

**evidence/example:** Research governance must balance autonomy with accountability, excellence with wider institutional capacity, mission-mode urgency with curiosity-driven work, national priorities with peer-review independence, and concentration in elite institutions with regional or state-university inclusion. A new apex institution does not remove these implementation trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Critically examine ANRF as a research-governance reform through autonomy, inclusion, mission...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Evaluate India's S&T institutional architecture using policy-status, programme-status and outcome-evidence discipline. Answer in about 300 words.

**Model thesis: Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** STI policy and Vigyan Dhara boundary. **Named evidence/example:** DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institution and status evidence ladder. **Named evidence/example:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.



### Claim -> named evidence -> analysis -> qualification:

- The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.
- The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology.
- DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments.
- Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner.

**Qualified conclusion: Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** STI policy and Vigyan Dhara boundary. **Named evidence/example:** DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institution and status evidence ladder. **Named evidence/example:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **evaluate** requires a direct position on "Evaluate India's S&T institutional architecture using policy-status, programme-status and...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

### Detailed examiner-grade model answer:

**Introduction and thesis: Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote

new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** STI policy and Vigyan Dhara boundary. **Named evidence/example:** DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institution and status evidence ladder. **Named evidence/example:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:**

State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** DST horizontal policy role. **Named evidence/example:** The Department of Science and Technology is a Department under the Ministry of Science and Technology, established in May 1971 to promote new areas and organise, coordinate and support science and technology activities. DST is a horizontal policy, programme and mission anchor, not India's single all-purpose laboratory network or a Ministry in its own right.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CSIR legal and

research-performing identity. **Named evidence/example:** The Council of Scientific and Industrial Research is an autonomous society registered under the Societies Registration Act and administratively under DSIR. Its official profile records a network of 37 national laboratories, 39 outreach centres, one Innovation Complex and three units across fields including chemicals, drugs, genomics, biotechnology, mining, aeronautics, instrumentation, environment and information technology. **Analysis:** This fixes the scientific process, system boundary, institution, application and

verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** STI policy and Vigyan Dhara boundary. **Named evidence/example:** DST's adopted-policy stack retains Science, Technology and Innovation Policy 2013 as the last clearly adopted national STI policy identified by the owners, while STIP 2020 was a draft and consultation exercise without a located adoption notification. Vigyan Dhara is a consolidated DST umbrella scheme; a policy, draft, umbrella scheme and individual grant are different instruments. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institution and status evidence ladder. **Named**

**evidence/example:** Constitution by law, notification, board meeting, programme launch, call for proposals, application, sanction, release, research activity, prototype, validation, licensing, commercialization, deployment and measured outcome are distinct evidence rungs. Counts, financing plans, current calls and policy status are date-sensitive and require an official dated owner. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Evaluate India's S&T institutional architecture using policy-status, programme-status and...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.